

ASSIGNMENT QUESTIONS

Q1. Variables & Data Types [5 Marks]

 Your Role

Cloud Intern @ AWS

 **Scenario:** You've just onboarded at AWS as an intern. Your manager gives you a CSV export of server metrics. Write an R script that stores the server name, CPU usage (%), RAM available (GB), and whether the server is online (TRUE/FALSE) in appropriately typed variables. Print each variable with a descriptive label.

 **Hint:** Use character, numeric, and logical data types.

 Your R Code / Answer:

Q2. R Syntax & Comments [5 Marks]

 Your Role

Junior SRE @ Google Cloud

 **Scenario:** Your SRE team uses R scripts for on-call runbooks. Write a well-commented R script that calculates the total disk usage across 5 virtual machines (values in GB). Comments should explain each step as if a teammate is reading the runbook during an incident at 3 AM.

 **Hint:** Use # for single-line comments. Prioritize clarity over cleverness.

 Your R Code / Answer:

Q3. Operators & Math [5 Marks]

 Your Role

FinOps Analyst @ Azure

 **Scenario:** Your team needs to calculate the monthly cloud bill. You have: base compute cost (\$4,200), storage cost (\$780), data egress cost (\$1,150), and a 12% discount coupon. Write an R script using arithmetic operators to compute the final bill after the discount. Also compute what percentage each category contributes to the pre-discount total.

 **Hint:** Use `+`, `-`, `*`, `/` and `%%` operators as needed.

 **Your R Code / Answer:**

Q4. Strings [6 Marks]

 Your Role

DevOps Intern @ Infosys Cloud

 **Scenario:** You are building an automated alert message system. Given a server name (e.g., 'prod-db-01'), region ('us-east-1'), and alert level ('CRITICAL'), use R string functions to: (a) convert the alert level to lowercase, (b) count the number of characters in the server name, (c) build a final alert string: '[CRITICAL] prod-db-01 in us-east-1 is DOWN'.

 **Hint:** Use `paste()`, `toupper()/tolower()`, `nchar()`, and `sprintf()` or `paste0()`.

 **Your R Code / Answer:**

Q5. If...Else [6 Marks]

 Your Role

Cloud Operations Intern @ TCS

 **Scenario:** You are writing a health-check script. Given a server's CPU usage value (input as a variable), write an R if-else chain that prints: 'Healthy' if CPU < 60%, 'Warning' if CPU is 60–85%, 'Critical' if CPU > 85%. Test your script with values 45, 72, and 91.

 **Hint:** Use `if / else if / else`. Run in a loop for the three test values.

 **Your R Code / Answer:**

Q6. While Loop [6 Marks] Your Role*SRE Intern @ Flipkart Cloud*

 **Scenario:** During a load test, your application retries a failed API call every 5 seconds. Simulate this in R: use a while loop that starts with attempt = 1, prints 'Attempt X: Retrying connection...' each iteration, and stops after 5 attempts (simulating max retries). On exit, print 'Max retries reached. Escalating to on-call engineer.'

 **Hint:** Use a counter variable and break condition inside the while loop.

 Your R Code / Answer:

Q7. For Loop [6 Marks] Your Role*Cloud Monitoring Intern @ Wipro*

 **Scenario:** You have a vector of 6 server response times in milliseconds: c(120, 340, 89, 450, 210, 95). Write a for loop that iterates over each value and prints: 'Server X: OK (Ym s)' if response < 200ms, or 'Server X: SLOW (Ym s)' if response >= 200ms. X is the server index (1–6), Y is the time.

 **Hint:** Use seq_along() or 1:length() to iterate with index.

 Your R Code / Answer:

Q8. Nested Loops [7 Marks] Your Role*Multi-Cloud Engineer Intern @ HCL*

 **Scenario:** Your company uses a 3x3 grid of data centers (3 regions x 3 availability zones). Write a nested for loop in R to print a connectivity check table: for each region ('us', 'eu', 'ap') and each zone ('z1', 'z2', 'z3'), print 'Checking us-z1... OK' etc. Simulate one failure: 'eu-z2' should print 'TIMEOUT' instead of 'OK'.

 **Hint:** Use outer and inner for loops with an if condition for the failure case.

 Your R Code / Answer:

Q9. Functions [8 Marks] Your Role

Cloud Automation Intern @ Accenture

 **Scenario:** Write a reusable R function called `check_uptime()` that takes two arguments: `total_minutes` (total time in a month) and `downtime_minutes`. The function should: calculate uptime %, print the SLA tier ('Gold' $\geq 99.9\%$, 'Silver' $\geq 99\%$, 'Bronze' $\geq 95\%$, 'Below SLA' otherwise), and return the uptime percentage. Test it with: (43200, 30), (43200, 300), (43200, 2500).

 **Hint:** Use `return()` and `if/else` inside the function. 43200 = minutes in a 30-day month.

 Your R Code / Answer:

Q10. Vectors [7 Marks] Your Role

Site Reliability Intern @ Razorpay

 **Scenario:** You collect CPU usage readings (%) every hour for 10 hours from a payment gateway server: `c(45, 52, 61, 78, 85, 90, 88, 72, 63, 55)`. Using R vector operations: (a) find the peak CPU hour (index), (b) find all hours where CPU exceeded 75%, (c) compute the average CPU usage, (d) add a new reading of 48% for hour 11 and print the updated vector.

 **Hint:** Use `which()`, `mean()`, `max()`, and `c()` for appending.

 Your R Code / Answer:

Q11. Lists [7 Marks]

 Your Role

Cloud Support Engineer Intern @ IBM Cloud

 **Scenario:** Create an R list called `incident` that stores a support ticket with the following fields: `ticket_id` ('INC-2024-001'), `severity` (2), `description` ('Database connection pool exhausted'), `affected_services` (a vector: 'auth-service', 'payment-service'), `resolved` (FALSE). Access and print each field individually. Then update `resolved` to TRUE and print the updated list.

 **Hint:** Use `list()`, `$` operator for access, and `[[ ]]` for indexing.

 **Your R Code / Answer:**

Q12. Data Frames [8 Marks]

 Your Role

Cloud Data Analyst Intern @ Jio Cloud

 **Scenario:** You receive a weekly resource utilization report. Create an R data frame with 5 rows representing 5 virtual machines, with columns: `vm_name`, `cpu_usage (%)`, `ram_used_gb`, `status` ('running'/'stopped'). (a) Display only VMs with `cpu_usage > 70%`. (b) Add a new column `cost_per_hour` (assume \$0.05 per % CPU). (c) Print the data frame sorted by `cpu_usage` descending.

 **Hint:** Use `data.frame()`, `subset()` or `df[condition,]`, and `order()`.

 **Your R Code / Answer:**

Q13. R Statistics – Mean, Median, Max/Min [7 Marks]

 Your Role

Performance Engineer Intern @ Myntra Tech

 **Scenario:** Your team is analyzing API latency data (in ms) collected over 12 hours: `c(145, 132, 189, 210, 98, 345, 122, 167, 201, 155, 143, 178)`. Calculate: (a) mean latency, (b) median latency, (c) max and min latency, (d) the difference between the max and min (latency range). Write a brief 2-line comment interpreting what high latency range means for a production system.

 **Hint:** Use `mean()`, `median()`, `max()`, `min()`.

 **Your R Code / Answer:**

Q14. Percentiles & Statistics [8 Marks] Your Role

Observability Intern @ Swiggy Cloud

 **Scenario:** In SRE, the P95 and P99 latency are critical SLO metrics. You have response times (ms) for 20 API calls. Generate a sample using: `set.seed(42); latency <- round(rnorm(20, mean=200, sd=50))`. Then compute: (a) P50, P90, P95, P99 using `quantile()`, (b) print whether P95 exceeds your SLO threshold of 350ms, (c) visualize the distribution using a simple histogram with a title 'API Latency Distribution'.

 **Hint:** Use `quantile()` with `probs` argument. Use `hist()` for the plot.

 Your R Code / Answer:

Q15. Visualization – Bar Chart [10 Marks] Your Role

Cloud BI Intern @ Tata Communications

 **Scenario:** You are preparing a monthly infrastructure cost report for your manager. Given the following monthly cloud costs by service: Compute=\$5200, Storage=\$1800, Networking=\$950, Database=\$2300, Security=\$600 — create a labeled bar chart in R with: (a) appropriate title 'Monthly Cloud Cost Breakdown', (b) colored bars (use `col=` with at least 3 colors), (c) x-axis label 'Cloud Service', (d) y-axis label 'Cost (USD)', (e) values displayed on top of each bar using `text()`.

 **Hint:** Use `barplot()`, `text()`, and the `height` argument for value labels.

 Your R Code / Answer: